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A multi-task deep learning network for robust and high accuracy star sensor image centroiding

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Abstract
Star sensors require reliable onboard centroid estimation despite binary-star interference and non-Gaussian distortions. Conventional algorithms lack robustness, while deep learning models often sacrifice accuracy or computational efficiency for real-time operation. This study proposes a lightweight multi-task CNN based on the HorNet baseline for onboard deployment. It simultaneously distinguishes single-star from binary-star targets and estimates sub-pixel centroids in a single forward pass. Two improvement plug-in modules—differentiable Gaussian convolution for noise suppression and coordinate attention for spatial enhancement—add minimal overhead, enhancing the centroid estimation accuracy for star spots of various morphologies. A Pareto-optimal loss function balances classification and regression without manual tuning. Experiments on a dataset (including ideal Gaussian, real non-Gaussian profiles, and binary-star interference) show that the proposed model achieves 100.0% binary-star detection rate, substantially outperforming conventional algorithms (≈53.7% accuracy). Under simulated non-Gaussian conditions, it attains a centroid error of ≈0.017 pixels, outperforming the conventional centroid method (≈0.16 pixels) and Gaussian fitting (≈0.12 pixels). The model remains compact, efficient, and easy to train, offering a structurally efficient solution with demonstrated potential for high-precision, real-time processing stellar centroid estimation on current GPU platforms, demonstrating potential for future onboard deployment.

Keywords Star sensor · Centroid estimation · Binary-star interference · Non-Gaussian distortion · Multi-task light CNN

Introduction

Star sensors operate in orbit using "window tracking" mode. Based on prior attitude information, known stars are projected onto the detector image, and small windows are generated around these projected positions (Fig. 1). A field-programmable gate array (FPGA) extracts these windows and processes them in parallel to estimate the centroid of each observed star. Under ideal conditions, each window contains exactly one stellar spot with a near-Gaussian intensity profile. These centroids are then used to determine the attitude of the sensor[1–3]. High-precision stellar centroiding is important not only for spacecraft attitude control but also for space physics, because

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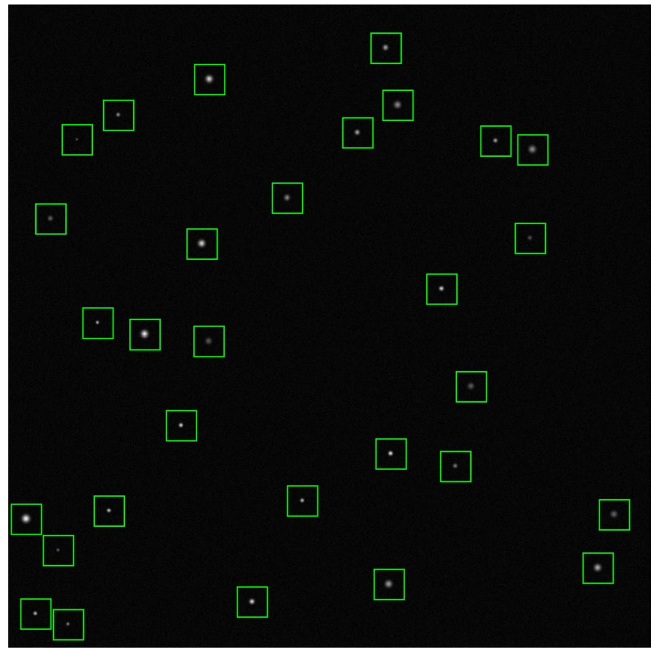

Fig. 1 A schematic diagram of FPGA window extraction from star images collected by a detector.

accurate attitude determination ensures the correct geolocation of magnetospheric plasma measurements, ionospheric imaging, and related observations.

Conventional centroiding algorithms can be broadly divided into two categories. The first category comprises sub-pixel localization methods[4–7], which are frequently based on grayscale information, such as the centroid method. These methods regard the centroid as an extremum point of the target, offer computational simplicity, but remain susceptible to spatial periodic errors and constrained localization accuracy. The second category employs fitting-based approaches, in which the stellar spot is fitted to a Gaussian or parabolic surface[6,8]. These methods achieve higher accuracy when the spot profile is approximately Gaussian, but they impose greater computational cost and exhibit substantial accuracy degradation when the profile deviates from this assumption. Consequently, FPGAs usually adopt the first category to satisfy real-time processing requirements.

Conventional algorithms have two principal limitations[9,10]. First, binary stars may fall within a single window. With the rapid expansion of internet constellations, more objects pass near stellar images, increasing the likelihood that two stars occupy the same window. Conventional methods identify such cases by examining each window through background thresholding and connected-component analysis. If a binary star is detected, the corresponding window is rejected to prevent large centroiding errors. However, when two stars are closely spaced, connected-component analysis may merge them into a single component, producing a large centroid error. Second, optical aberrations and dominant Gaussian detector noise distort stellar spots away from the ideal Gaussian profile. Conventional algorithms lack integrated noise suppression or irregular-profile fitting capabilities; therefore, accurate estimation of the true centroid remains difficult[11,12].

On-orbit embedded processors have advanced rapidly, enabling the use of GPUs and deep learning for real-time spaceborne image processing[13–15]. Unlike conventional methods, deep learning models capture rich spatial and semantic information by learning pixel-level representations across multiple layers. For example, the European Space Agency's Phi-Sat missions successfully demonstrated onboard artificial intelligence[16]. Lightweight neural networks have been developed for cloud detection, image classification, and target recognition[17,18]. The field has further expanded to hyperspectral analysis, disaster monitoring, and planetary terrain modeling[19,20].

Recently, deep learning has been applied directly to star-sensor processing. Dave, et al.[21] proposed RSOnet for classifying stellar points and tracking space objects. Fang, et al.[22] employed a UNet-based network[23] to suppress noise and estimate centroids. Casetti-Dinescu, et al.[24] developed a VGG-based model[25] for stellar-spot centroiding. Zapevalin, et al.[14] employed an artificial neural network to estimate centroids from windowed star images. Although recent studies[11,12] have demonstrated the potential of deep learning for star centroiding under specific conditions, their models either require substantial computational resources, such as large transformer architectures, or omit binary-star classification. In contrast, the proposed deep learning model addresses both classification and regression for on-orbit deployment, achieving higher accuracy under non-Gaussian distortions while maintaining real-time inference capability.

Processing a windowed star image therefore involves two tasks: classification, namely single-star versus binary-star identification, and regression, namely centroid estimation for single stars. A deep learning approach can process many windows in parallel, analogous to FPGA-based processing. However, most existing deep learning methods[22–24] rely on relatively simple Convolutional Neural Networks (CNNs) and have been evaluated on limited datasets. The stellar-spot images in those studies were obtained from high-quality telescopes[14] and did not include severe distortion or low signal-to-noise ratio (SNR). Although these methods outperformed conventional algorithms in some comparisons, their centroiding accuracy remained below the 0.02-pixel sub-pixel requirement.

Early CNNs such as VGG and ResNet[26] are not well suited to numerical regression and often fail to satisfy the accuracy requirements of stellar-spot centroiding[24]. Transformers[27] provide higher fitting accuracy but are computationally large and slow. Recent studies have combined transformers with CNNs to improve accuracy while retaining compact and efficient models[28,29]. Object detection frameworks such as YOLO[30,31] and CenterNet[32] can also perform classification and centroiding, but they are unsuitable for this task. These methods are designed for images containing multiple objects from different categories. They require preprocessing, including anchor boxes and heatmaps, and post-processing, including non-maximum suppression, none of which is required in the proposed method. Moreover, they perform multiple regressions, including width, height, and center estimation, simultaneously, and conflicts among these outputs reduce center-estimation accuracy. Other lightweight, high-accuracy CNNs[29,33] have been proposed for real-time processing.

The objective of this study is to develop a deep learning method for windowed star images that achieves higher fitting accuracy than existing approaches while overcoming the limitations of conventional algorithms. Specifically, this study presents a methodological framework for star-image processing as a technical report on sensor and data analysis in space physics. The main contributions are as follows: (1) a multi-task CNN that outputs both classification and centroiding results in one forward pass, thereby offering a structural reduction in theoretical computational overhead compared with two separate single-task CNNs, as both tasks share the same feature extraction backbone in a single forward pass; (2) a differentiable Gaussian convolution (DGC) layer for adaptive denoising, combined with a coordinate attention (CA) mechanism for enhancing spatial features and further improving centroiding accuracy; (3) a Pareto-optimal loss optimization strategy that balances classification and regression tasks so that each achieves accuracy comparable to that obtained when trained independently; and (4) a method for generating windowed star-image datasets that include Gaussian, non-Gaussian, and binary-star images with various brightness levels, in-window positions, and SNR values, thereby covering all targeted testing conditions.

Materials and methods

Overall architecture of the multi-task deep learning model

The proposed multi-task deep learning model, MultiTask Net (MTNet), is based on a CNN architecture. In this framework, the feature map output by the CNN after image processing is flattened into a one-dimensional vector, which is then passed through two fully connected layers to produce two groups of predictions representing classification probability and centroid coordinates. This model structure is illustrated in Fig. 2.

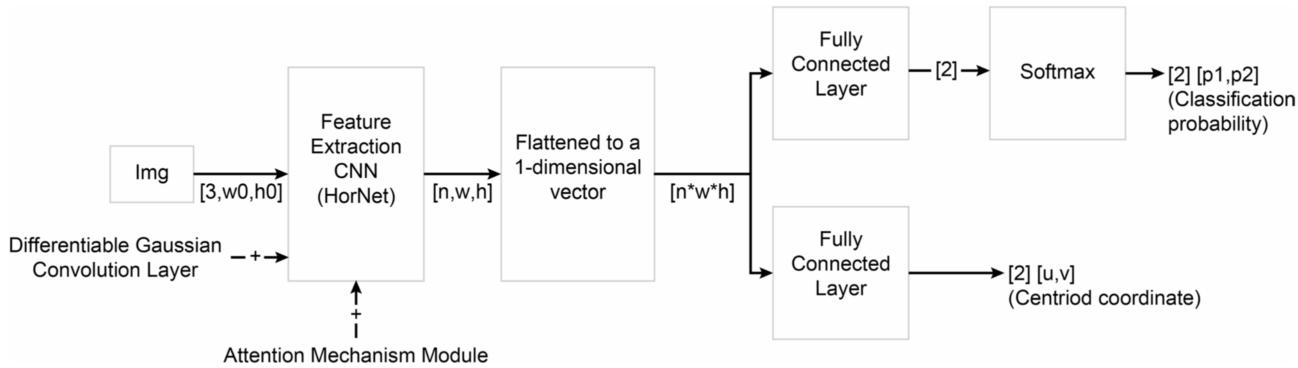

Fig. 2 A schematic diagram of the overall MTNet model architecture.

In conventional CNN architectures, multiple layers are connected in series to form blocks, which are stacked repeatedly. Multiple blocks can then be connected in series to form a stage for use in processing feature maps without changing the resolution. CNNs can be constructed by stacking individual stages, with deeper networks consisting of higher stage counts.

In this study, a feature-extraction CNN was first selected from among recently developed, high-accuracy, lightweight CNNs, including the three network architectures discussed below. (1) The DLA[33] network provides an aggregation structure to achieve deeper information fusion. This framework is based on early CNN models (e.g., ResNet), but was modified by introducing aggregation modules to fuse feature-map information across different stages, as well as feature information from various blocks within the network. Experiments have shown that this process, which combines the advantages of feature pyramid networks (FPNs), substantially improves fitting accuracy[33]. (2) The ConvNext[29] network combines the complementary strengths of a transformer, which can capture global image features, with a CNN, which can capture local features and exhibits translation invariance. This strategy draws on the Swin-Transformer (S-T) architecture[28], which modifies early CNN models by replacing batch normalization with layer normalization and incorporating operations such as depthwise convolutions and inverted bottlenecks from the MobileNet series[34,35]. This implementation also involves weighted summation operations, similar to those in a multi-head attention mechanism. The resulting ConvNext model achieves classification accuracies that exceed S-T, applied to extensive datasets[29], requiring the same number of FLOPs while exhibiting significantly lower model complexity. (3) The HorNet[36] model suggests that, due to nonlinearity, higher-order spatial interactions (between two spatial positions in deep model feature maps) can improve model fitting capabilities. Similarly, the recursive gated convolution module (g^nConv) further facilitates interactions between higher-order spatial information in specific feature maps, on the basis of a design adopted by ConvNext. The resulting fitting accuracy surpassed that of both S-T and ConvNext with the same number of FLOPs[36]. This principle, utilized by the g^nConv module, is detailed in the original study[36]. In this case, a higher order enables more sufficient interactions among spatial positions in the feature map (i.e., performing more vector dot product calculations.) but also results in greater computational costs.

A comparison of these implementation principles suggested that HorNet is an improvement over ConvNext (i.e., g^nConv is added based on the latter model). Notably, DLA emphasizes information fusion among feature maps with different resolutions, while HorNet places greater emphasis on the fusion of spatial positional information within feature maps. Given that centroiding principles are based on the spatial positions of star-spot pixels, HorNet was selected as the feature-extraction backbone. Experiments are also presented in which HorNet was replaced by DLA and ConvNext. Comparisons are made in terms of model size, inference speed, and accuracy detailed in the "Feature-extraction model replacement" section.

Proposed model improvements

While the fitting accuracy achieved by the original CNN model (i.e., HorNet) is primarily represented by the centroid coordinates of star spots, a gap remains when compared with the ideal centroiding accuracy attainable by conventional Gaussian fitting algorithms (detailed in the Comparative experiments with conventional algorithms section). Considering the characteristics of windowed star images, this study proposes two improvements to further enhance the fitting accuracy of the original CNN. (1) Differentiable Gaussian convolution layer to adaptively remove noise interference. (2) CA mechanism module to enhance spatial feature extraction.

Differentiable Gaussian convolution layer

Since noise in real on-orbit star images is dominated primarily by Gaussian noise, removing noise interference and improving star-image quality are conducive to centroiding[12]. As such, this study develops and implements a differentiable Gaussian convolution layer (DGC), which parameterizes and learns the Gaussian filtering used in conventional image processing, integrates the results into the proposed CNN, and thereby enabling adaptive feature smoothing that suppresses noisy activations while preserving spatially relevant star-spot patterns.

The DGC layer performs Gaussian convolutions on input feature maps using a kernel that can be expressed as:

$$\mathrm{G}\left(\mathrm{x},\mathrm{y}\right)=\left(\frac{1}{2\pi\,\sigma^{\,2}}\right)\times\,\exp\left(-\frac{x^2+y^2}{2\sigma^{\,2}}\right),$$

where (x,y) represents the coordinates of each parameter and σ denotes the standard deviation. In this case, only σ needs to be a learnable parameter, while all other parameters can be calculated from the standard deviation. During training, σ is continuously updated and the Gaussian kernel is recalculated in each forward propagation. Different values of σ then correspond to distinct filtering and smoothing effects. The size of convolution kernels is determined by the image size and must be odd to ensure symmetry of the Gaussian distribution. Larger images correspond to larger convolution kernels, to improve Gaussian filtering effects. Based on practical experience, if image size is below 10×10, it is set to 3×3. If the size is above 60×60, it is set to 7×7.

When a feature map of shape [B, C, H, W] is convolved in a given layer, a different Gaussian kernel is set for each channel in the feature map using a different learnable parameter σ (C in total). During training, each channel can independently and adaptively learn and adjust σ, thereby improving the effects of Gaussian convolution. In this study, the final feature map output by the feature-extraction CNN is processed by a DGC and the edges are padded to preserve feature-map resolution. Zero padding with the kernel's size//2 is adopted to preserve spatial resolution. The discrete Gaussian kernel is explicitly renormalized as:

kernel = kernel/kernel.sum() ,.

after each σ update to prevent gain variation. The learnable σ is parameterized in log-space to ensure positive, initialized to 5.05, clamped to [0.1, 10.0]. For each channel, the discrete Gaussian kernel is normalized independently over its spatial dimensions, ensuring that the sum of each channel's kernel weights equals 1.0. This channel-wise independent normalization prevents the overall feature magnitude from varying with the learned σ parameters. These configurations are applied during both training and inference. This step was included because the final-layer feature map contains the richest semantic feature information and the largest channel dimensions, making it more suitable for separate processing of different channels and enabling better convolutional denoising and smoothing effects.

Attention mechanism module

The included attention mechanism module enabled the lightweight CNN to further learn to focus on important pixel information in images while suppressing pixel information that should be ignored. In this study, for example, star spot pixels represent important information, while background and noise should be suppressed to improve the fitting accuracy. The earliest attention mechanism module, Squeeze-Excitation Net (SE-Net)[37], enables CNNs to focus on important channels while suppressing unimportant channels. This module first employs global average pooling to obtain average values for elements in each channel. However, subsequent studies[38] have shown that this operation causes a loss of critical spatial information in the feature map. For example, the positions of pixels in star spot images have a significant influence on classification and centroiding tasks. To compensate for this deficiency, attention mechanism modules, such as the convolutional block attention module (CBAM)[38], use large-scale convolution kernels to acquire spatial information. However, this method cannot capture information between long-range elements and therefore embeds spatial details into channel information by improving the first step of SE-Net[13]. As a result, the input feature map undergoes global average pooling separately along the width and height directions, followed by subsequent parallel processing. In this way, spatial information can be acquired between long-range elements in a given feature map. Extensive experiments have demonstrated that the coordinate attention (CA) module[13] is more effective than other modules (e.g., CBAM) for improving CNN accuracy, while also incurring significantly lower computational costs[13]. Notably, CA achieved the best performance and was used to process the final feature map output by the feature-extraction CNN, while keeping all feature map dimensions unchanged. Experiments were conducted to swap the usage order of the DGC module and the CA module, and the results indicated that applying the DGC module to the final feature map before the CA module produced better results (data not shown).

Model loss function calculation method

In this study, ground truth values used for classification tasks were designed with one-hot encoding, forming binary arrays in which the first two elements were 1 and 0 (Category 1) or 0 and 1 (Category 2). The predicted binary array was subsequently processed by a Softmax layer and the predicted value was taken at the position where the corresponding ground-truth element equaled 1. Loss for that sample was then calculated using the cross-entropy loss function.

In centroiding tasks, the ground truth for a single-star sample is the true (u, v) coordinate value. In this study, u and v were each expressed as their ratios to the window width and height, respectively, to eliminate the influence of different window sizes and to increase generalizability. For example, given a window size of 24×24 and a centroid coordinate of (12.5, 12.0), the ground truth would be labeled as (0.52083, 0.5). In contrast, ground truth arrays for binary-star samples were set as (0.0, 0.0). The neural network then predicted a binary array for each sample. In the case of images whose corresponding ground truths were a single star, the two predicted values and the ground truth were used to calculate MSELoss. For images with a corresponding binary star ground truth, the loss was set to 0 by default and the images were not involved in loss calculation. During inference, if the classification branch predicts a binary star, the regression output is discarded and excluded from attitude determination.

Optimized design of the model training loss function

When loss is calculated and backpropagated in a multi-task network, the training optimization objective comprises a linear combination of multiple tasks when there is no competition. During backpropagation using the loss function, losses for individual tasks can be summed and averaged to obtain a surrogate loss. However, when competitive relationships exist among tasks, decreases in loss constrain each loss term and the weights of task losses must be balanced. Otherwise, one or more task losses may fail to decrease during the optimization and iteration steps[39]. The case studied in this study involved competition between two tasks of the proposed model. Notably, when the loss weights were set to 0.5, the final accuracy achieved by the centroiding task (after training) differed substantially from the final accuracy achieved when the model was configured and trained as a single-task model with centroiding only (detailed in the Optimizing the model loss function section).

Previous research on methods for allocating task-loss weights in multi-task training have often treated multi-task learning as a multi-objective optimization problem[40]. The overall objective is to find a Pareto-optimal solution that ensures the reduction of one task loss term in multi-task learning does not adversely affect the reduction of other task losses. Since this process requires relatively little computation, relative to performance levels, it has been widely applied in the field of deep learning. The primary implementation involves using a shared feature map Z in the last layer of the multi-task neural network, which in this study is a one-dimensional vector obtained by flattening the final output feature map in the feature-extraction CNN.

For each task, the gradient information generated by backpropagating through the feature map Z is used in an interaction calculation to obtain the weight proportion that should be assigned to each task loss. In the two-task case considered here, the loss weight for the first task is given by:

$$a = \left[\frac{(\nabla_z L2 - \nabla_z L1)^T \nabla_z L2}{\|\nabla_z L2 - \nabla_z L1\|^2} \right],$$

where L1 and L2 denote losses for the first and second tasks, respectively, and $\nabla_z L1$ and $\nabla_z L2$ denote gradients obtained by backpropagating the L1 loss and the L2 loss to Z, respectively. The range of a should then be constrained to [0, 1] and the loss weight for the second task is denoted $(1-a)$.

Dataset simulations

The training and test datasets used in this study included ideal Gaussian-profile star spots, non-Gaussian-profile star spots, and binary-star window images, each with a resolution of 24×24 pixels. The dataset generation process is shown in Fig. 3, where (u_i, v_i) denotes the ground-truth label for the windowed star-spot coordinates.

Ideal Gaussian star spots. These spots were simulated using the Photutils package. Each stellar spot was assigned a stellar magnitude, representing its total energy, and the corresponding energy distribution was generated by sampling a two-dimensional Gaussian function. The ground-truth centroid corresponds to the center of this function.

Non-Gaussian star spots. These star spots were generated using optical simulation software (CODEV 2025), incorporating optical-system aberrations with varying levels of severity. A complete optical model of the star sensor was established in CODEV, and ray tracing was employed to simulate the propagation of point light sources located at different field positions through the optical system, thereby generating PSFs that incorporated realistic aberrations, including spherical aberration, coma, and astigmatism, as well as diffraction effects. The resulting images exhibited energy and brightness distributions that were consistent with the imaging characteristics of real instruments[41]. After propagation through the optical system, rays affected by optical aberrations formed non-Gaussian star spots on the image plane. The ground-truth centroid was defined as the image-plane position of the chief ray.

Binary-star images: Two star-spot coordinates were randomly generated within each image window. To characterize the binary-star test set, we recorded the following parameters: (1) the separation distance, which was uniformly sampled from 1 to 22 pixels; (2) the flux ratio, which was determined from stellar magnitudes independently assigned within the range of 0 to 10 mag; (3) the SNR, which ranged from 6 to 25 dB. At each coordinate, either a Gaussian or non-Gaussian star spot was simulated. Non-Gaussian star spots were generated using CODEV. These images have no ground-truth centroid.

Examples of Gaussian star-spot images are shown in Fig. 4 below, including differences in both star-spot profiles and image signal-to-noise ratios. Figure 5 shows non-Gaussian star-spot images, which illustrate the profile characteristics of non-Gaussian star spots. Binary-star windowed images are shown in Fig. 6, which demonstrates different positional relationships between the two stars.

A total of 200,000 images were generated for each of the Gaussian and non-Gaussian single-star sets, with stellar magnitudes corresponding to 0–10. Another 18,000 binary-star windowed images were generated covering

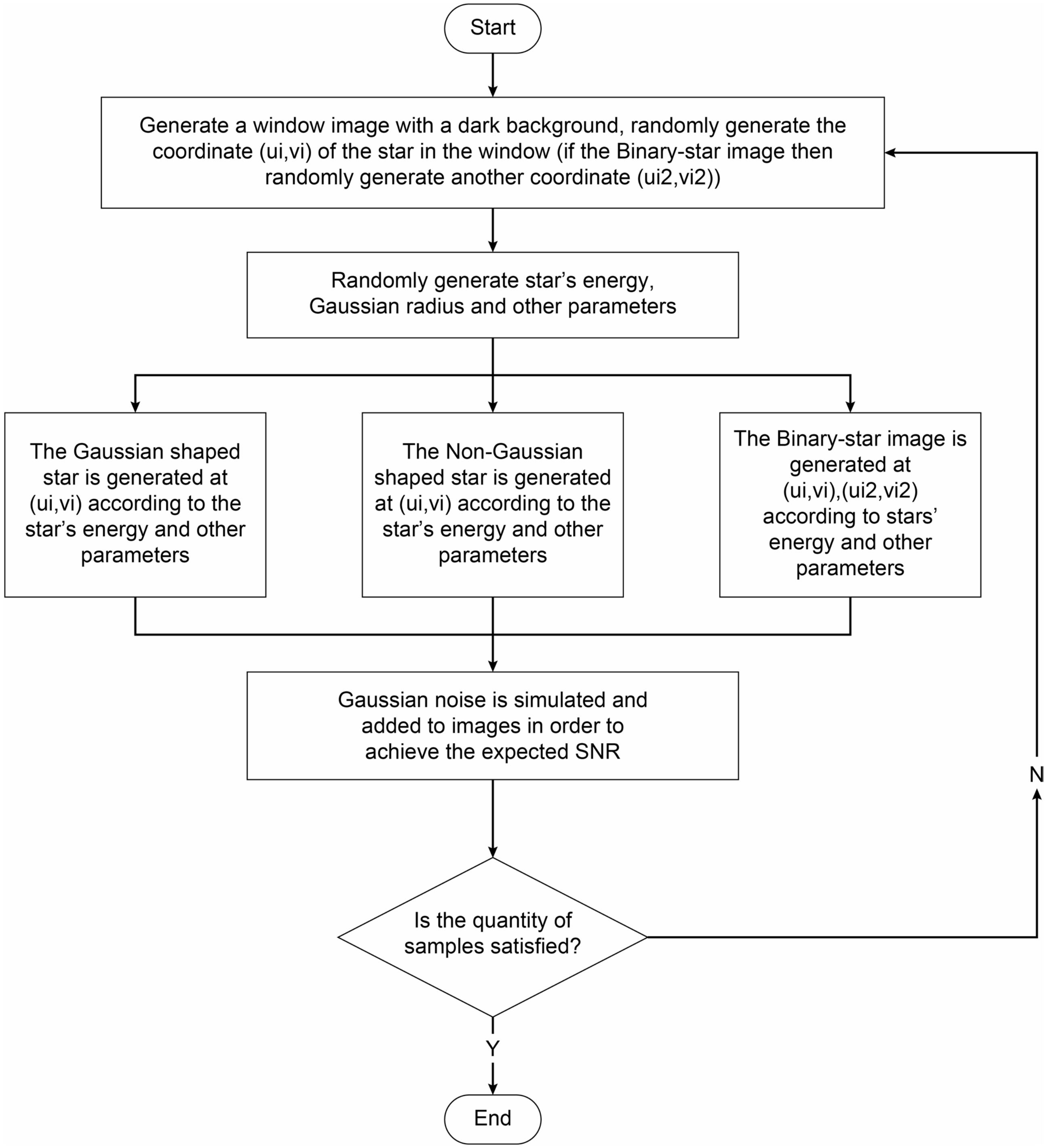

Fig. 3 A flowchart for generating single-star and binary-star window images.

all positional relationships. Random-intensity Gaussian noise was added to all images. Distortion with random degree of severity was added to all non-Gaussian star images.

The single-star images were split into training (180,000), validation (10,000), and test (10,000) sets. The binary-star images were equally divided into three portions and mixed into each set. The validation dataset was used for synchronous verification in each training epoch. After model parameters were updated, predictions were made in the validation dataset to observe changes in errors between predicted values and ground truths, to

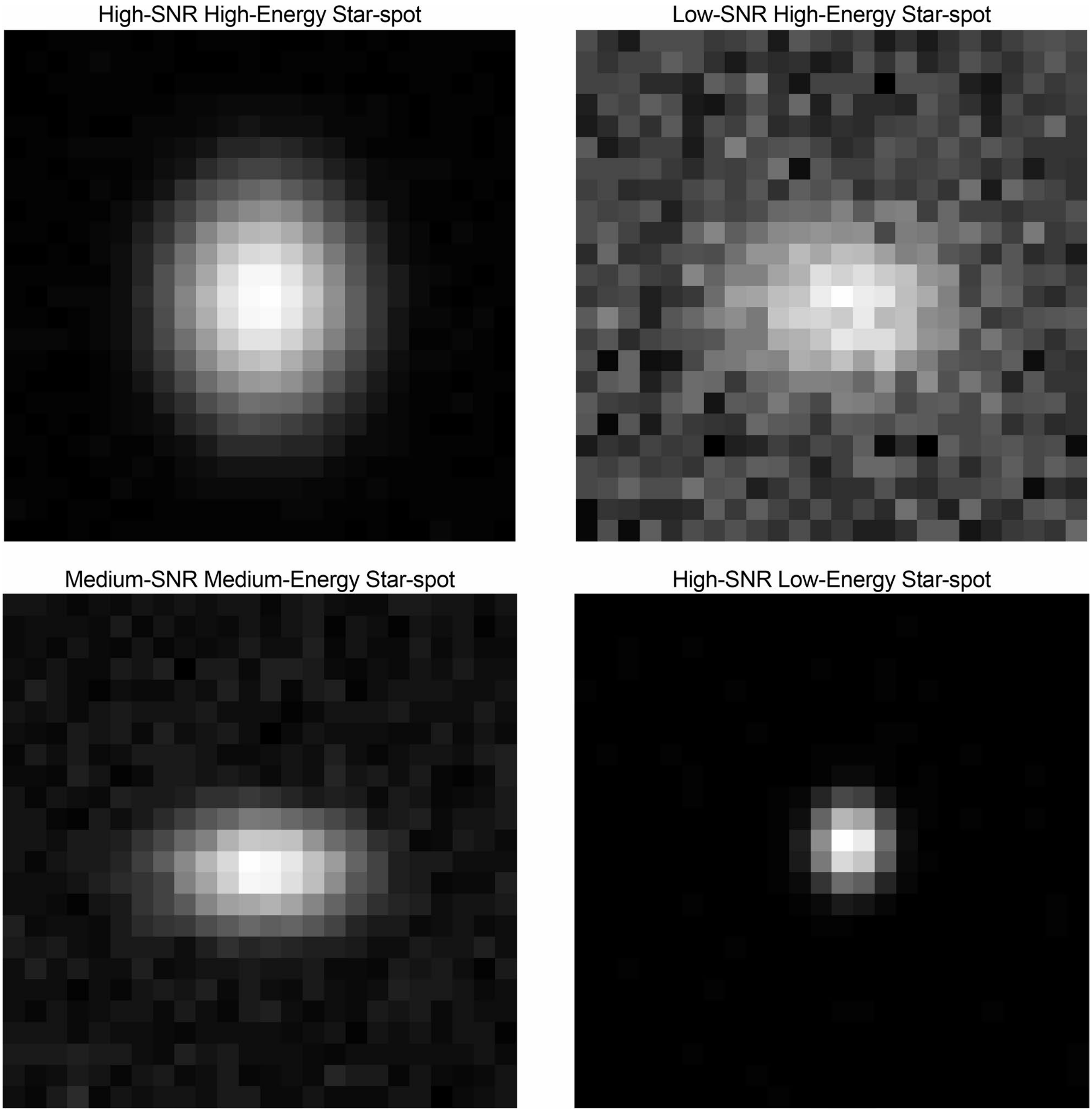

Fig. 4 Gaussian star-spot window images.

determine model convergence, stop training, and prevent overfitting. The test dataset was used to evaluate accuracy for comparison with diffrent algorithms. Binary-star samples are oversampled to 37.5% in validation/test sets (vs. 3.2% in training) for statistical reliability.

Implementation details

Network: The network adopts the HorNet variant "hornet_tiny," with depths = [2, 3, 18, 2] and base_dim = 64. Each 24×24 input image is resized to 96×96 using bicubic interpolation and converted into a three-channel

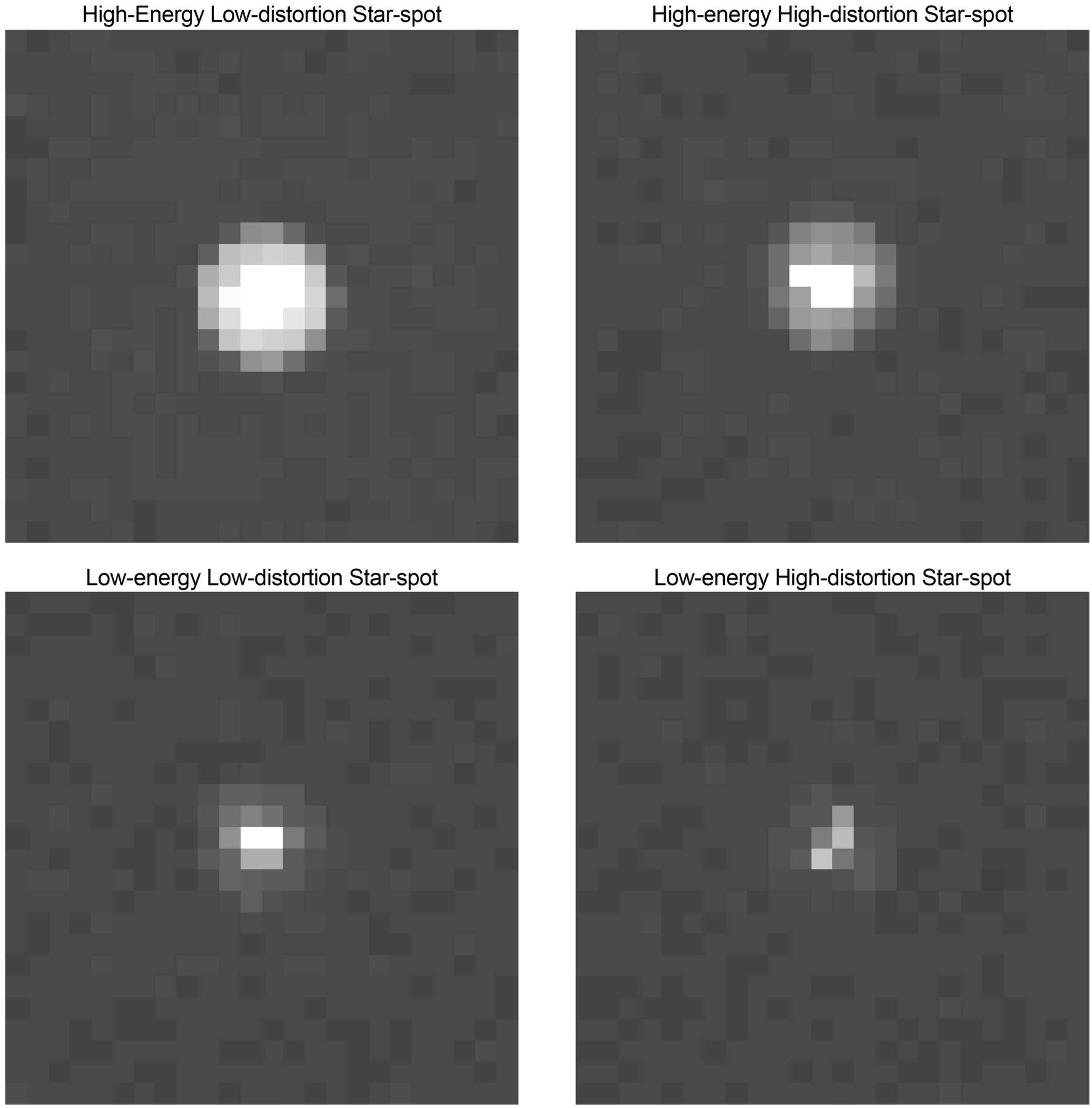

Fig. 5 Non-Gaussian star-spot window images.

image. The final feature map has dimensions of $512 \times 6 \times 6$. The CA module is applied after the fourth stage, whereas the DGC module is applied before the CA module to the final feature map generated by the fourth stage.

Training: The model is optimized using Adam with an initial learning rate of 1×10^{-4}, which is reduced to 1×10^{-5} at epoch 40 and to 1×10^{-6} at epoch 80. The batch size is 64, the weight decay is set to 0, and Kaiming uniform initialization is employed. The model is trained for 100 epochs, and the checkpoint corresponding to the epoch with the lowest centroid error on the validation set is retained.

Data generation: The centroid position is sampled uniformly within each 24×24 window; the magnitude ranges from 0 to 10; the background standard deviation ranges from 2 to 5 DN; the background mean ranges from 50 to 200 DN; and the SNR ranges from 6 to 25 dB. Non-Gaussian star spots are generated using a CODEV

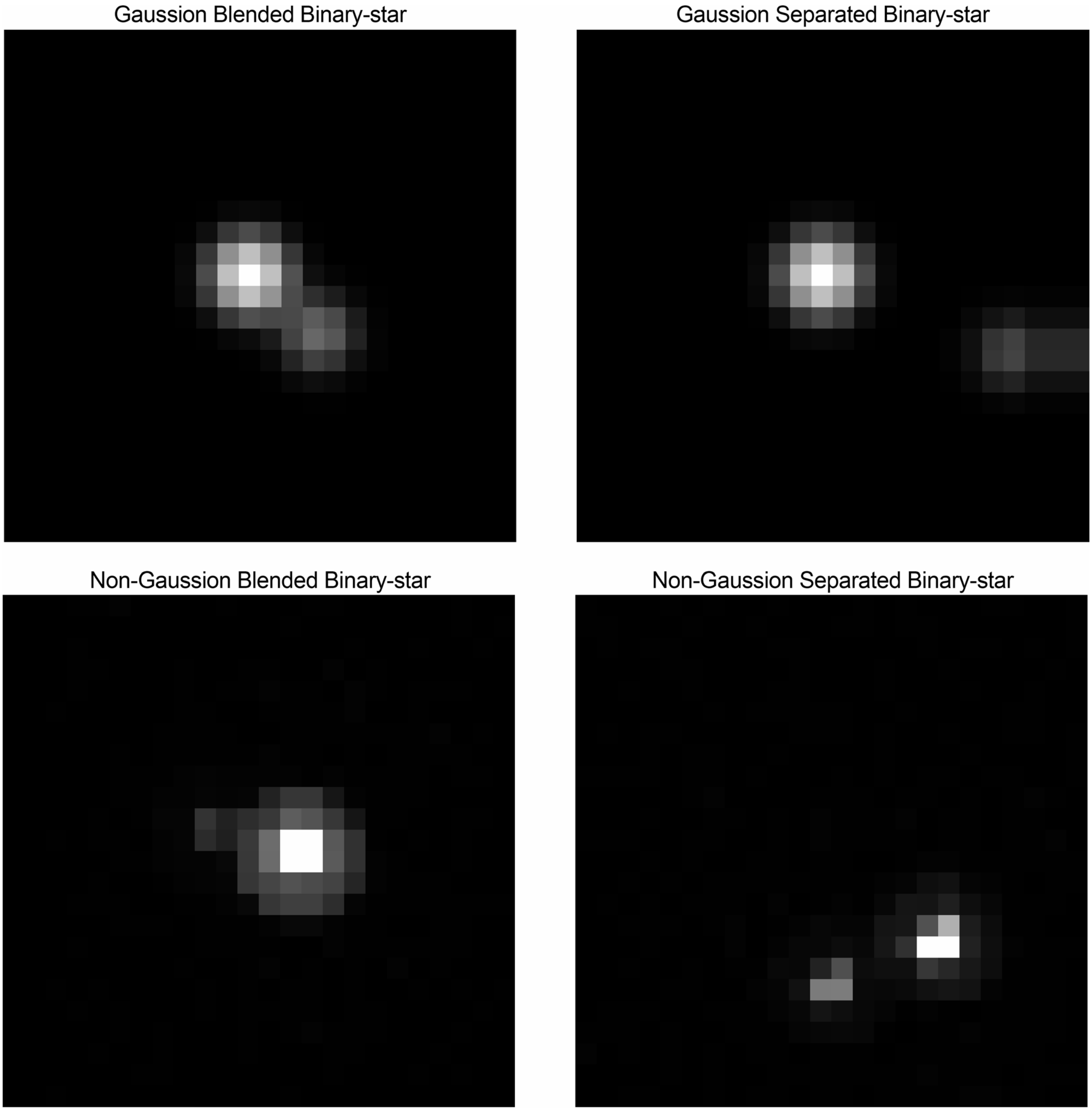

Fig. 6 Binary-star window images.

model configured with the field of view of a real on-board star sensor, and the light source is modeled as black-body radiation with a color temperature of 6000 K. The total energy of each star spot $F(m)$ is calculated by its magnitude m:

$$F(m) = F_0 \cdot 10^{-0.4\,m},$$

where F_0 is the total energy corresponding to $m = 0$. The reference F_0 is calculated based on the specific detector configuration of the star sensor. The definitions of SNR are shown below:

$$SNR_{lin} = \frac{I_{\text{peak}} - \mu_{\text{bg}}}{\sigma_{\text{bg}}},\ SNR_{\text{dB}} = 20\log_{10}(SNR_{lin}),$$

where I_{peak} is the maximum pixel value of the star spot (before adding noise), μ_{bg} is the mean background intensity (before adding noise), and σ_{bg} is the standard deviation of the additive Gaussian noise (background noise).

Results

Four experiments were conducted to verify that the fitting accuracy of MTNet surpasses that of conventional algorithms and to confirm the effectiveness of the proposed improvements, as described in the subsections below.

Evaluation metrics. Detection accuracy is defined as the proportion of binary-star samples correctly identified (binary-star detection rate), i.e., Detection accuracy = TN/(TN + FP), where "binary-star" is designated as the negative class (label = 0) and "single star" as the positive class (label = 1) in the implementation. The decision threshold is set to 0.5; specifically, a sample is classified as a binary star when the predicted probability of the single-star class is less than 0.5. The standard deviation (std), measured in pixels, defined as the discrepancy between estimated centroids and their corresponding ground-truth centroids, is used to measure the centroiding errors accuracy.

Datasets. The non-Gaussian dataset was used to evaluate performance under realistic conditions, whereas the ideal Gaussian dataset was used for comparison with conventional algorithms under near-ideal conditions (the Comparative experiments with conventional algorithms section).

Comparative experiments with other algorithms

Conventional algorithms used in this study misclassify binary stars as single stars. Binary-star detection rate is used to measure their detection accuracy. Centroiding results from misclassified binary stars are excluded to focus on single-star performance for conventional algorithms. All conventional baseline methods were tuned using the validation set. Centroid method: The star-extraction threshold was defined as the mean background intensity within the image window plus a tunable coefficient multiplied by the corresponding background standard deviation; this coefficient was set to 3.3 based on the SNR range of the window images and empirical considerations. Gaussian fitting: Considering the star-spot dimensions and the image SNR range represented in the dataset, the fitting window was set to 7 × 7 pixels, with an initial σ value of 2.5. Optimization was performed using the Levenberg–Marquardt algorithm, and the stopping criterion was defined as a parameter-change tolerance of 1×10^{-9} to improve centroid estimation accuracy. Connected-component method: The minimum connected-component area was set to 4 pixels, whereas the maximum area was limited to 50% of the image-window area. In addition, the VGG-based method[24] was reimplemented and evaluated using identical dataset partitions and evaluation metrics. Comparative results for the primary evaluation metrics are shown in Table 1 below. Statistical error curves for single-star centroiding coordinates (u, v), calculated from the test dataset, are shown in Fig. 7 (ideal Gaussian condition) and Fig. 8 (Non-Gaussian condition), where Δu is the statistical error curve for u and Δv is the statistical error curve for v. Table 2 reports an expanded set of metrics of MTNet: signed bias, MAE, RMSE, the 50th and 95th percentiles of radial error, the outlier rate (radial error > 0.1 pixel), 95% confidence interval (based on the normal approximation), and the erroneous-output rate attributable to misclassified binary-star samples. Collectively, these metrics demonstrate that the model provides reliable centroid estimates for the majority of evaluated cases.

Definitions for Table 2 metrics are shown below:.

Let N = number of samples, W = window size (here 24 pixels), For the i-th sample:.

$$\mathrm{e_{u,i}} = (\mathrm{u_{pred,i}} - \mathrm{u_{gt,i}}) \cdot \mathrm{W},$$

Table 1 A comparison with conventional and other deep-learning algorithms.

Star-spot conditions	Method	Binary-star detection rate	Centroiding std (pixel)
Ideal Gaussian condition	MTNet	100.0%	(0.0172, 0.0173)
	VGG-based[24]	100.0%	(0.0212, 0.0208)
	Centroid method	—	(0.0326, 0.0326)
	Gaussian-fitting method	—	(0.0166, 0.0166)
	Conventional classification	53.7%	—
Non-Gaussian condition	MTNet	100.0%	(0.0175, 0.0173)
	VGG-based[24]	100.0%	(0.0796, 0.0787)
	Conventional centroiding	—	(0.1598, 0.1597)
	Gaussian-fitting centroiding	—	(0.1219, 0.1217)
	Conventional classification	53.7%	—

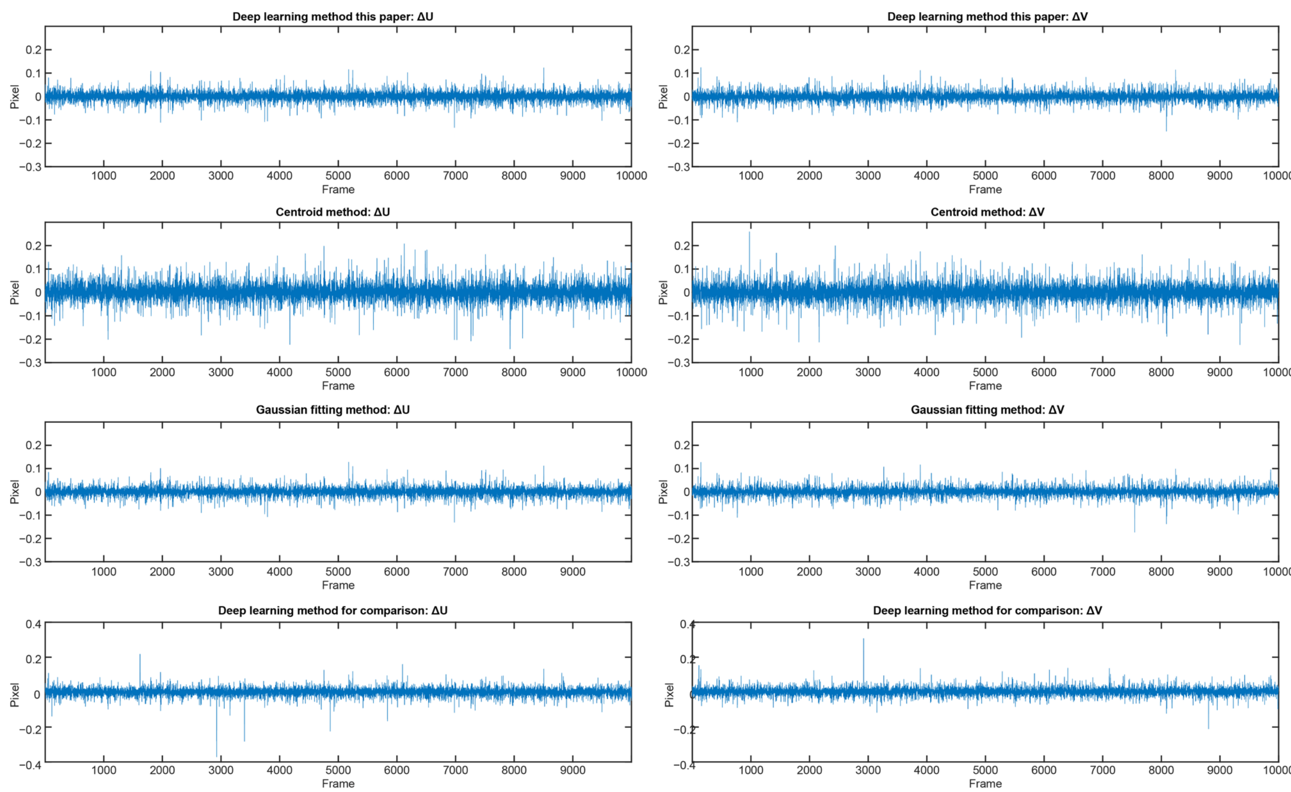

Fig. 7 Statistical error curves for centroid coordinates under ideal Gaussian conditions.

$$e_{v,i} = (v_{pred,i} - v_{gt,i}) \cdot W,$$

$$r_i = \sqrt{e_{u,i}^2 + e_{v,i}^2},$$

where $e_{u,i}$, $e_{v,i}$, r_i represent the u-direction, v-direction, and radial error, respectively. Using the u-direction as an example:.

$$Bias_u = \frac{1}{N}\sum_{i=1}^{N} e_{u,i},$$

$$Std_u = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N} (e_{u,i} - Bias_u)^2},$$

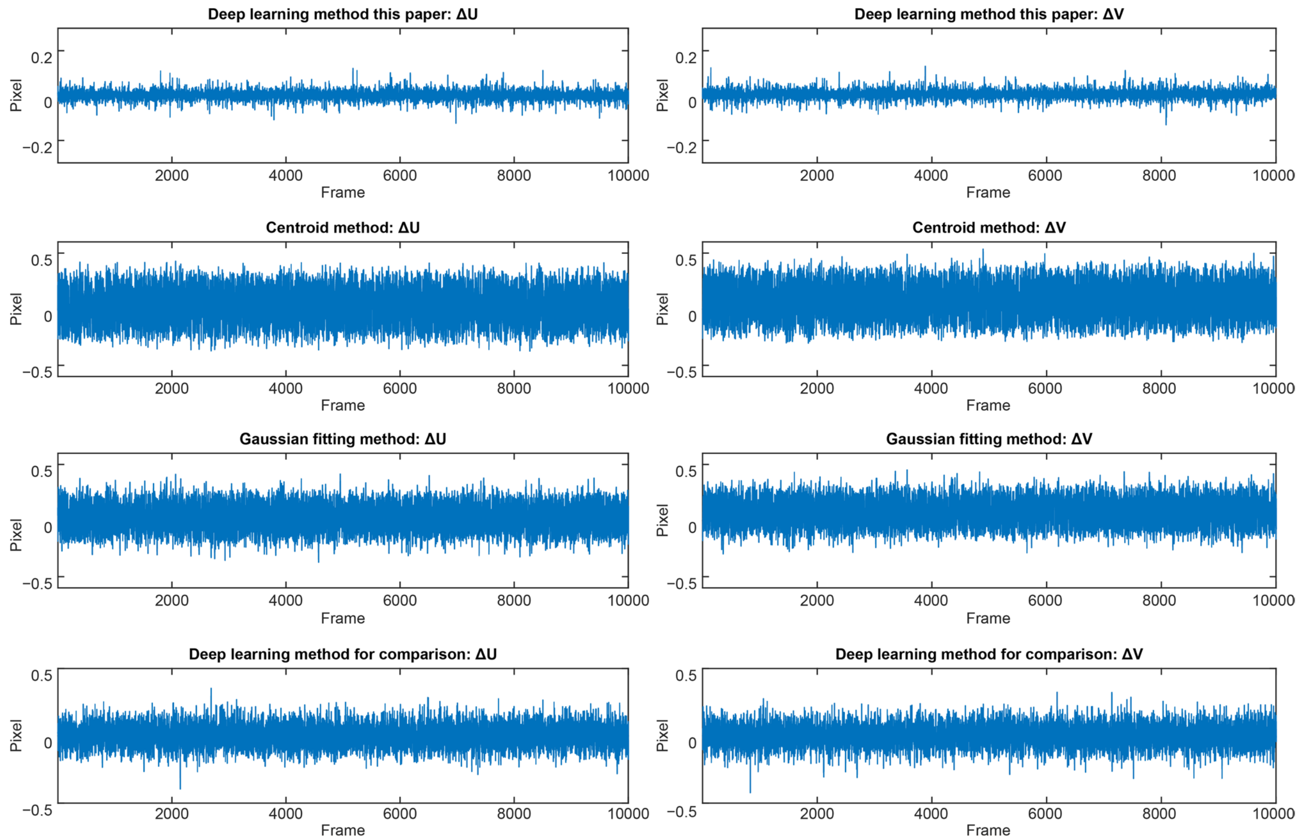

Fig. 8 Statistical error curves for centroid coordinates under non-Gaussian conditions.

Table 2 Extended centroid error statistics for the proposed MTNet under Gaussian and simulated Non-Gaussian conditions.

Metric	Gaussian Condition	Non-Gaussian Condition
u-direction Bias (mean) (pixel)	−0.0015	0.0012
v-direction Bias (mean) (pixel)	0.0007	0.0050
u-direction Std	0.0172	0.0175
v-direction Std	0.0173	0.0173
u-direction MAE (pixel)	0.0128	0.0125
v-direction MAE (pixel)	0.0125	0.0131
u-direction RMSE (pixel)	0.0179	0.0175
v-direction RMSE (pixel)	0.0176	0.0180
Radial Error Median (50th) (pixel)	0.0160	0.0164
Radial Error 95th Percentile (pixel)	0.0495	0.0499
Outlier Rate (>0.1 pixel)	0.21%	0.22%
95% Confidence Interval for Radial Mean	[0.0198, 0.0204]	[0.0199, 0.0205]
Error Output Rate (Misclassified Binary Stars)	0.0%	0.0%

$$\mathrm{MAE_u} = \frac{1}{N}\sum_{i=1}^{N} |e_{u,i}|,$$

$$\mathrm{RMSE_u} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_{u,i}^2},$$

$$\mathrm{Median}(r) = r_{\left(\frac{N+1}{2}\right)} \quad (\text{after sorting } r_i \text{ ascending}),$$

$$P_{95}(r) = r_{(\lceil 0.95N \rceil)} \quad (\text{the } \lceil 0.95N \rceil - \text{th smallest value}),$$

$$\mathrm{OutlierRate} = \frac{1}{N}\sum_{i=1}^{N} I_{\{r_i>0.1\}},$$

where $I_{\{r_i>0.1\}}$ is the indicator function.

$$\mathrm{CI}_{95\%} = \left[\bar{r} - 1.96\frac{s_r}{\sqrt{N}},\ \bar{r} + 1.96\frac{s_r}{\sqrt{N}}\right] \ (\text{using normal approximation}),$$

where $\bar{r} = \frac{1}{N}\sum_{i=1}^{N} r_i$, $s_r = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(r_i - \bar{r}\right)^2}$.

MTNet and achieve 100.0% binary-star detection rate (Visually, the point-spread morphology resulting from the blending of binary stars is readily distinguishable from that of a single star; therefore, it should also be easily discriminable by CNNs), far better than conventional algorithm (53.7%). The confusion matrix obtained by MTNet on the test set was TN = 6000, FP = 0, FN = 0, and TP = 10,000. These results demonstrate a binary-star detection rate of 100.0%, a false-positive rate for binary stars of 0.0%, and a single-star recall of 100.0%. Evaluation on the mixed dataset comprising Gaussian and non-Gaussian binary-star samples also achieved a classification accuracy of 100.0%, suggesting that the model identifies the global pattern associated with binary-star overlap rather than relying on PSF-specific morphological characteristics. By comparison, the confusion matrix obtained by the conventional algorithm on the test set was TN = 3222, FP = 2778, FN = 0, and TP = 10,000, corresponding to a binary-star detection rate of 53.7%, a false-positive rate for binary stars of 46.3%, and a single-star recall of 100.0%.

Furthermore, the results of classification performance of the proposed deep learning method and the conventional algorithm on binary-star samples of varying difficulty levels within the test dataset are summarized as follows. For challenging binary-star samples (115 images) characterized by close centroid separation (< 5 pixels), large magnitude difference (> 5 mag), and low SNR (< 10 dB), the proposed method correctly identified all cases as binary stars, whereas the conventional algorithm achieved a correct identification rate of 1.73%. For moderately difficult binary-star samples (502 images) with moderate separation (5 to 12 pixels), small magnitude difference (< 5 mag), and high SNR (> 15 dB), the proposed method maintained 100% correct identification, while the conventional algorithm correctly identified 66.33% of the samples. For easy binary-star samples (402 images) with large separation (> 14 pixels), small magnitude difference (< 5 mag), and high SNR (> 18 dB), both the proposed method and the conventional algorithm achieved 100% correct identification.

Under Gaussian conditions, The centroiding accuracy of MTNet is much higher than the centroid method and only 0.0006 pixels worse than Gaussian fitting (which is near the ideal limit). Under non-Gaussian conditions, Gaussian fitting deteriorates sharply, while MTNet maintains high accuracy, demonstrating robust performance under the simulated star-spot conditions. In summary, MTNet outperformed conventional algorithms used to compare with in the present study, in terms of both detection and centroiding accuracy during the processing of star-sensor windowed images. Furthermore, the experimental results demonstrate that the proposed deep learning method achieves higher centroiding accuracy than the previously reported VGG-based method[24].

The results presented in Table 2 demonstrate that MTNet achieves high centroid estimation accuracy under both Gaussian and simulated non-Gaussian conditions. The signed biases along the u and v directions are negligible

Table 3 Comparative results from an ablation experiment with CA only.

Star-spot conditions	Accuracy	With CA	Without CA
Ideal Gaussian condition	Binary-star detection rate	100.0%	100.0%
	Centroiding std (pixel)	(0.0172 ± 0.0003, 0.0173 ± 0.0003)	(0.0178 ± 0.0004, 0.0176 ± 0.0004)
Non-Gaussian condition	Binary-star detection rate	100.0%	100.0%
	Centroiding std (pixel)	(0.0175 ± 0.0003, 0.0173 ± 0.0003)	(0.0191 ± 0.0004, 0.0188 ± 0.0004)

Table 4 Comparison results for the ablation experiment with DGC only.

Star-spot conditions	Accuracy	With DGC	Without DGC
Ideal Gaussian condition	Binary-star detection rate	100.0%	100.0%
	Centroiding std (pixel)	(0.0172 ± 0.0003, 0.0173 ± 0.0003)	(0.0177 ± 0.0003, 0.0177 ± 0.0003)
Non-Gaussian condition	Binary-star detection rate	100.0%	100.0%
	Centroiding std (pixel)	(0.0175 ± 0.0003, 0.0173 ± 0.0003)	(0.0192 ± 0.0003, 0.0195 ± 0.0003)

(< 0.005 pixels), indicating that the model exhibits little estimation bias. Under both conditions, the median radial errors remain below 0.02 pixels, while the 95th-percentile radial errors are approximately 0.05 pixels, confirming that the majority of centroid estimates satisfy the subpixel accuracy requirement. The outlier rates for radial errors exceeding 0.1 pixels are approximately 0.2%, indicating stable performance with only a minimal proportion of large-error cases. The binary-star recall of 100.0% corresponds to a zero erroneous-output rate attributable to misclassified binary stars, ensuring that classification failures do not introduce centroid deviations. Collectively, these complementary metrics provide robust evidence that MTNet delivers reliable centroid estimation performance across diverse star-spot conditions.

Ablation and ordering study with two proposed improvements

Ablation experiments were conducted separately for CA alone, DGC alone, and with both proposed improvements removed, to verify the different degrees of influence for CA and DGC when processing images with different star-spot profiles. This was done under two conditions: ideal and non-Gaussian-profile single-star windowed star images. CA and DGC were respectively added to and removed from MTNet during training and testing. All ablation results are consistently reported as the mean ± standard deviation across five independent runs, with each model retrained and evaluated separately, and the standard deviations rounded to four decimal places. Furthermore, the results of the DGC–CA ordering experiment confirm that applying DGC before CA achieves higher centroid estimation accuracy than applying the modules in the reverse order. Comparisons of inference results for the test dataset are described below.

1. *Ablation experiments with CA only*:

Table 3 demonstrates that removing CA did not affect detection accuracy. In the centroiding task, CA improved model performance under both star-spot conditions, with improvements being more pronounced in non-Gaussian conditions. This outcome confirms the effectiveness of the CA attention mechanism for improving the fitting accuracy of MTNet.

2. *Ablation experiment with the differentiable Gaussian convolution layer only*:

Table 4 demonstrates that removing DGC did not affect detection accuracy. In the centroiding task, DGC improved model performance under both star-spot conditions, with improvements being more pronounced under non-Gaussian conditions. This outcome confirms the effectiveness of DGC for improving the fitting accuracy of MTNet.

Table 5 Comparative results from an ablation experiment with both CA and DGC removed.

Star-spot conditions	Accuracy	With Both	Without Both
Ideal Gaussian condition	Binary-star detection rate	100.0%	100.0%
	Centroiding std (pixel)	(0.0172 ± 0.0003, 0.0173 ± 0.0003)	(0.0180 ± 0.0004, 0.0179 ± 0.0004)
Non-Gaussian condition	Binary-star detection rate	100.0%	100.0%
	Centroiding std (pixel)	(0.0175 ± 0.0003, 0.0173 ± 0.0003)	(0.0263 ± 0.0004, 0.0266 ± 0.0004)

Table 6 The quantitative results of the DGC-CA ordering experiment.

Configuration	Star-spot conditions	Binary-star detection rate	Centroid-ing std (pixel)
DGC before CA	Ideal Gaussian condition	100.0%	(0.0172 ± 0.0003, 0.0173 ± 0.0003)
	Non-Gaussian condition	100.0%	(0.0175 ± 0.0003, 0.0173 ± 0.0003)
CA before DGC	Ideal Gaussian condition	100.0%	(0.0309 ± 0.0005, 0.0285 ± 0.0005)
	Non-Gaussian condition	100.0%	(0.0426 ± 0.0005, 0.0468 ± 0.0005)

Since DGC and CA are inserted at the same location within the model and differ only in order; parameter count and computational cost remain unchanged between the two configurations.

3. *Ablation experiment with both CA and the differentiable Gaussian convolution layer removed*:

Table 5 demonstrates that removing both modules did not affect detection accuracy. However, in the centroiding task, the combined use of the two modules produced a more significant model improvement (under both star-spot conditions). For example, under non-Gaussian conditions, compared with using only one of the modules, the centering standard deviation is further reduced by approximately 0.0016 pixels. While removing both led to a clear decline in fitting accuracy. This result verified the effectiveness of the combined use of CA and DGC for improving the overall fitting accuracy of MTNet.

4. *Ordering experiment with CA and the differentiable Gaussian convolution layer*:

Table 6 shows that applying DGC before CA achieves higher centroiding accuracy than applying the modules in the reverse order. The performance degradation observed in the CA-before-DGC configuration can be attributed to a cascading error mechanism. When CA is applied to noise-contaminated feature maps, its attention weights are derived from noisy activations. Consequently, the module may amplify noise peaks while suppressing genuine star-spot signals. This multiplicative reweighting irreversibly distorts the feature map before it is processed by DGC. Although DGC can attenuate high-frequency noise, it cannot recover informative signals that have already been suppressed by inaccurate attention weighting. This mechanism explains why the CA-before-DGC configuration performs worse than either module used individually, as the two modules counteract each other when arranged in this order. In contrast, the DGC-before-CA configuration follows a more effective processing sequence, in which denoising precedes spatial feature enhancement, thereby yielding the best performance.

Optimizing the model loss function

A two-task loss-weight allocation method was adopted while training the MTNet under the non-Gaussian conditions described in Experiment 1. Loss-weights are updated at every training batch, and loss-weights values are clipped to the [0, 1] interval for stability safeguard; the computation graph is retained for gradient aggregation. Weight values for the two tasks calculated by the proposed method in each epoch are shown in Fig. 9, where the number of training epochs was set to 100. Throughout the training process, adapting calculated weights assigned to the classification-task loss were relatively small (~0.0001), while the weights assigned to the centroiding-task loss were relatively large (close to 1.0).

To further validate the effectiveness of the proposed adaptive weighting strategy, we conducted additional experiments involving: (1) statically assigned loss weights, including the default setting of 0.5/0.5 and the optimal

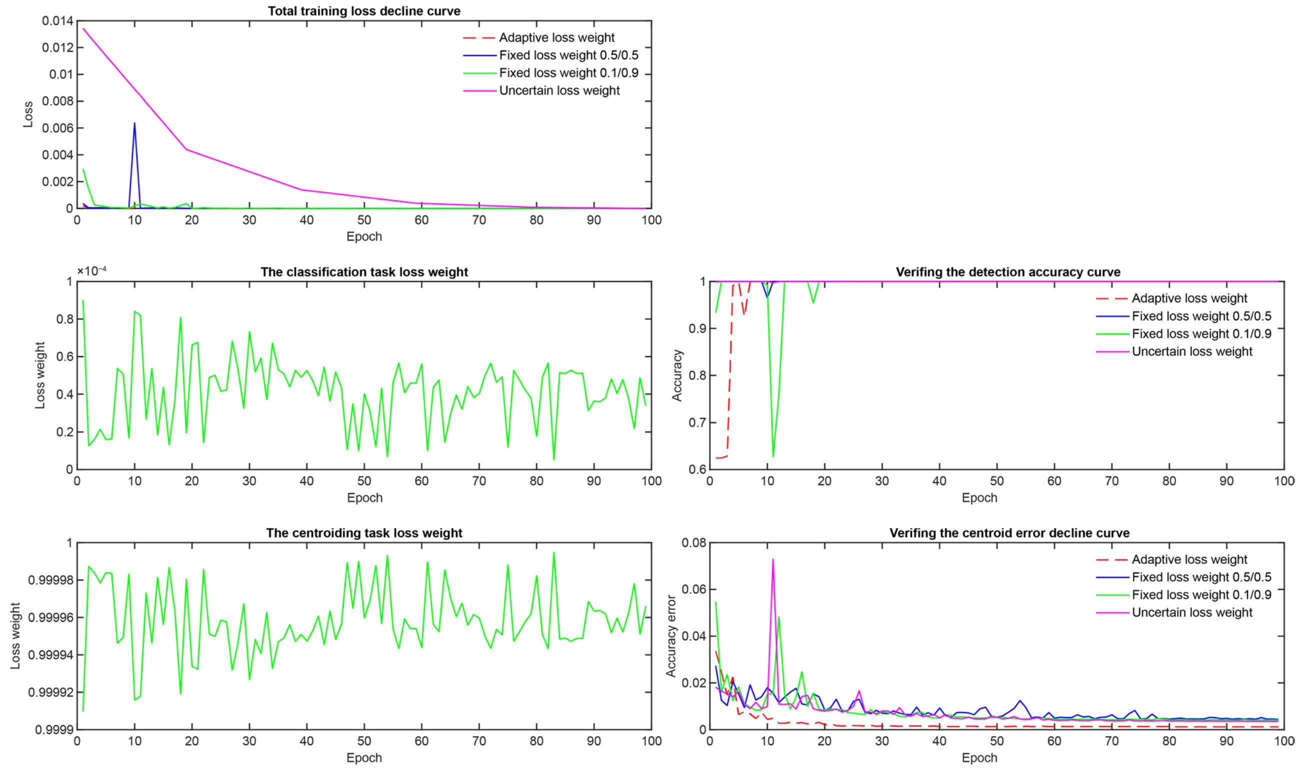

Fig. 9 Variation curves for the two task-loss weights used during training, which provides a comparison of training performance with fixed loss weights and Uncertain loss weight.

ratio of 0.1/0.9 identified through a grid search over {0.1/0.9, 0.3/0.7, 0.7/0.3, 0.9/0.1}, with the detailed grid-search procedure omitted for conciseness; and (2) the uncertainty weighting method[39], hereafter abbreviated as "uncertain loss weight."

The model was then retrained for comparisons, Fig. 9 shows epoch-wise descending curves for the total training loss under these two settings, as well as variation curves for detection accuracy and centroiding error obtained through synchronous validation during training. It is evident that the final detection accuracy converges to 100.0% in all cases. The adapting loss calculating method proposed in this study produced a final centroid MAE error (the mean of the radial errors between all predicted values and their corresponding ground truth values, which is computed on normalized coordinates (range [0, 1]) and is therefore dimensionless) of 0.00117, and a MAE error of 0.00115 on the test data set; which was essentially the same as the final converged centroiding error (0.00111 (validation), 0.00110 (test)), produced when the model was trained only on the centroiding task, with classification-task loss weights set to zero. In contrast, when both loss weights were fixed at 0.5, the final MAE error converged to 0.00439 (validation) and 0.00432 (test). When both loss weights were fixed at 0.1/0.9, the final MAE error converged to 0.00411 (validation) and 0.00410 (test). The uncertain loss weight produced final MAE errors of 0.00355 (validation) and 0.00348 (test). The total loss values from different weighting schemes are not directly comparable because the loss functions differ in scale and composition; their curves are shown only to illustrate convergence behavior and are not used as a performance criterion. The final evaluation relies on the two task-specific metrics: detection accuracy and mean radial error (on normalized coordinates). This demonstrated that the proposed loss-function optimization method enabled both tasks to achieve optimal training results.

Table 7 Comparative results for the three CNNs.

CNNs	Star-spot conditions	Binary-star detection rate	Centroid-ing std (pixel)	Params Size (MB)	FLOPs (G)	Peak Memory (MB)	Latency (ms/64 imgs)
HorNet	Ideal Gaussian condition	100.0%	(0.0172, 0.0173)	22.15	0.73	6002.36	12.75
	Non-Gaussian condition	100.0%	(0.0175, 0.0173)				
DLA	Ideal Gaussian condition	100.0%	(0.0184, 0.0181)	22.18	0.80	5601.70	12.95
	Non-Gaussian condition	100.0%	(0.0207, 0.0201)				
ConvNext	Ideal Gaussian condition	100.0%	(0.0176, 0.0174)	28.45	0.82	5890.57	12.37
	Non-Gaussian condition	100.0%	(0.0192, 0.0191)				

Feature-extraction model replacement

The feature-extraction CNN in MTNet was replaced by DLA and ConvNext, as training and testing were conducted under the same hardware platform environment (Hardware: NVIDIA RTX 4090 D, AMD Ryzen 9 9950X, 128GB RAM; latency measured on desktop GPU). The trained models were then compared with the original model in terms of model size, prediction speed, and prediction accuracy, as shown in Table 7. The comparative experiments described above suggest that when the feature-extraction CNN in the model is replaced by DLA or ConvNext, the resulting centroiding accuracy is inferior to that of HorNet. Regarding computational efficiency, HorNet exhibits the lowest FLOPs and the smallest model size among the three candidate backbones, while its peak memory usage is only slightly higher than those of the other two models and its inference latency remains comparable to those of DLA and ConvNeXt. Given its improved fitting accuracy and competitive computational cost, HorNet was selected as the feature-extraction backbone of the proposed MTNet.

Discussion

Experiments with conventional algorithms demonstrate that both single-star centroiding precision and binary-star classification accuracy were unsatisfactory: nearly half of binary-star images were misclassified, and centroiding accuracy under non-Gaussian conditions failed to meet sub-pixel requirements. In contrast, the proposed deep learning method avoids these bottlenecks (lacking the optimization mechanisms for noise suppression and profile fitting). It achieves 100.0% binary-star detection rate on anomalous star images under the tested conditions and fits centroid coordinates with much higher precision than conventional algorithms meeting sub-pixel requirements. The multi-task model, built on a single feature-extraction network, processes both tasks simultaneously per image, reducing computational overhead. Two proposed improvements effectively boost centroiding accuracy. First, a differentiable Gaussian convolution (DGC) layer applies learnable smoothing to deep feature maps, which helps suppress noisy activations and enhance feature representations relevant to centroid localization. Second, the coordinate attention (CA) mechanism — one of the lightest plug-and-play attention modules that integrate channel and spatial information—enhances CNN fitting accuracy. Additionally, Pareto-optimal loss optimization balances the two tasks, enabling each to achieve accuracy comparable to that obtained through single-task training. The overall model remains lightweight and straightforward to train, with architectural characteristics that are favorable for potential onboard deployment. However, establishing practical flight feasibility requires additional hardware-specific optimization, model quantization, and validation on real onboard platforms.

The binary-star classification accuracy of 100.0% was achieved using a simulated dataset under the specific experimental conditions considered in this study. Real-world PSF anomalies, complex stellar blending scenarios, and unmodeled noise sources may generate edge cases that are not represented in the current simulation framework.

The reported inference latency indicates the potential for real-time processing. However, these measurements were obtained using a desktop GPU and therefore do not represent the performance achievable on flight-qualified processors, such as FPGAs or radiation-hardened GPUs. Establishing practical on-board feasibility requires hardware-specific optimization, model quantization, and validation on flight-representative edge-computing platforms. Therefore, the deployment claim should be interpreted as an assessment of the architectural potential for real-time operation rather than as verified evidence of on-board capability.

The simulated dataset incorporates only additive Gaussian noise and does not represent the multiple noise sources encountered in actual on-orbit environments, including Poisson noise, dark-current noise, defective pixels, and transient bright spots induced by proton irradiation. These unmodeled factors may reduce the centroiding accuracy of the proposed model on real flight data relative to the performance obtained under simulated conditions. Consequently, the experimental findings should be regarded as preliminary validation under simplified noise conditions. Future work will incorporate more realistic mixed-noise models, validate the proposed method using real star sensor data, and address non-Gaussian noise by employing generative models such as U-Net to preprocess and further denoise star images, with the objective of improving centroiding accuracy and robustness to interference.

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Data availability The simulated single-star and binary-star windowed image datasets generated and analyzed in this study are not publicly archived yet but are available from the corresponding author upon reasonable request. The source code for MTNet can be obtained from the corresponding author (Y. Wu, 20033110168@stu.xidian.edu.cn) for the purpose of reproducing the results or building upon the reported research. All underlying data needed to understand, evaluate, and build upon the reported findings are presented in the paper.

Declarations

Competing interests The authors declare no competing interests.

References

1. Li, X., Liu, G. & Jinlin, N. Autofocusing of ISAR images based on entropy minimization. *IEEE Trans. Aerosp. Electron. Syst.* **35**, 1240–1252. https://doi.org/10.1109/7.805442 (1999).
2. Liebe, C. C. Accuracy performance of star trackers - a tutorial. *IEEE Trans. Aerosp. Electron. Syst.* **38**, 587–599. https://doi.org/10.1109/taes.2002.1008988 (2002).
3. Shuster, M. D. & Oh, S. D. Three-axis attitude determination from vector observations. *J. Guid Control*. **4**, 70–77. https://doi.org/10.2514/3.19717 (1981).
4. Shortis, M. R., Clarke, T. A. & Short, T. Comparison of some techniques for the subpixel location of discrete target images. In *SPIE Proceedings.* 239–250 (SPIE, Boston, MA, United States, 1994).
5. Stone, R. C. A comparison of digital centering algorithms. *Astron. J.* **97**, 1227. https://doi.org/10.1086/115066 (1989).
6. Wei, X., Zhang, G. & Jiang, J. Subdivided locating method of star image for star sensor. *J. Beijing Univ. Aeronaut. Astronaut.* **29**, 812–815. https://doi.org/10.3969/j.issn.1001-5965.2003.09.014 (2003).
7. Alexander, B. F. Elimination of systematic error in subpixel accuracy centroid estimation [also Letter 34(11)3347–3348(Nov1995)]. *Opt. Eng.* **30**, 1320–1331. https://doi.org/10.1117/12.55947 (1991).
8. Zhang, G., Wei, X. & Jiang, J. Full-sky autonomous star identification based on radial and cyclic features of star pattern. *Image Vis. Comput.* **26**, 891–897. https://doi.org/10.1016/j.imavis.2007.10.006 (2008).
9. Arimura Fialho, M. A. Evaluation of centroiding algorithms for an autonomous star tracker. *Eur. Phys. J. Spec. Top.* **234**, 7877–7883. https://doi.org/10.1140/epjs/s11734-025-01917-0 (2026).
10. Liang, B., Zhu, H., Zhang, T. & Tong, Y. Research status and development tendency of star tracker technique. *Chin. Opt.* **9**, 16–29. https://doi.org/10.3788/co.20160901.0016 (2016).
11. Zhao, H., Lembeck, M. F., Zhuang, A., Shah, R. & Wei, J. Real-time convolutional-neural-network-based star detection and centroiding method for CubeSat star tracker. *IEEE Trans. Aerosp. Electron. Syst.* **61**, 8172–8184. https://doi.org/10.1109/taes.2025.3542744 (2025).
12. Luo, Z., Guo, Q., Feng, J., Li, Y. & Zhang, X. Centroid algorithm for high-dynamic star sensor based on key point detection deep learning algorithms. *Opt. Express*. **33**, 17203–17232. https://doi.org/10.1364/oe.542489 (2025).
13. Hou, Q., Zhou, D. & Feng, J. Coordinate attention for efficient mobile network design. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR),* 13708–13717 (IEEE, Nashville, TN, USA, 2021).
14. Zapevalin, P. R., Novoselov, A. & Zharov, V. E. Artificial neural network for star tracker centroid computation. *Adv. Space Res.* **71**, 3917–3925. https://doi.org/10.1016/j.asr.2022.11.023 (2023).
15. Luo, Z., Feng, J., Guo, Q. & Li, Y. Centroid localization of dynamic star sensors under proton irradiation. *Opt. Precis Eng.* **33**, 2089–2107 (2025).
16. NASA. *How NASA is testing AI to make earth-observing satellites smarter* (2025). https://www.nasa.gov/science-research/earth-science/how-nasa-is-testing-ai-to-make-earth-observing-satellites-smarter/
17. Chanoui, M. A., Khalil, I., Sbihi, M., Alaoui Ismaili, Z. E. A. & Guennoun, Z. Trends and applications of on-board image processing for earth observation nanosatellites: A systematic review. *Int. J. Aeronaut. Space Sci.* **26**, 1989–2019. https://doi.org/10.1007/s42405-025-00885-y (2025).
18. Duggan, A., Andrade, B. & Afli, H. Advancing Earth observation: a survey on AI-powered image processing in satellites. *Eur. J. Remote Sens.* **58**, 2567921. https://doi.org/10.1080/22797254.2025.2567921 (2025).
19. Jenisha, C. & Kalirajan, S. KV, A. D. Recent innovation of deep learning approches in satellite imagery: A comprehensive review. In *2024 International Conference on Sustainable Communication Networks and Application (ICSCNA)*. 1213–1220 (IEEE, Theni, India, 2024).
20. Sun, C. et al. Super resolution powered efficient and reliable satellite image data acquistion. In *Proceedings of the ACM SIGCOMM 2025 Conference,* 311–328 (ACM, New York, NY, USA, 2025).
21. Dave, S., Clark, R., Lee, R. S. K. & RSOnet An image-processing framework for a dual-purpose star tracker as an opportunistic space surveillance sensor. *Sensors* **22**, 5688. https://doi.org/10.3390/s22155688 (2022).
22. Fang, B. L., Wang, J. G. & Feng, G. B. Calculation of spot entroid based on physical informed neural networks. *Acta Phys. Sinica*. **71**, 200601. https://doi.org/10.7498/aps.71.20220670 (2022).
23. Ronneberger, O., Fischer, P. & Brox, T. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015* (eds Navab, N. et al.) 234–241 (Springer International Publishing, 2015).
24. Casetti-Dinescu, D. I., Girard, T. M., Baena-Gallé, R., Martone, M. & Schwendemann, K. Star-image centering with deep learning: HST/WFPC2 images. *Publ Astron. Soc. Pac.* **135**, 054501. https://doi.org/10.1088/1538-3873/acd080 (2023).
25. Simonyan, K. & Zisserman, A. Very deep convolutional networks for large-scale image recognition. In *3rd International Conference on Learning Representations (ICLR*), 1–14 (San Diego, CA, USA, 2015). (2015).
26. He, K., Zhang, X., Ren, S. & Sun, J. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR),* 770–778 (IEEE, Las Vegas, NV, USA, 2016).

27. Vaswani, A. et al. Attention is all you need. In *Proceedings of the 31st International Conference on Neural Information Processing Systems,* 6000–6010 (Curran Associates Inc., Red Hook, NY, USA, 2017).
28. Liu, Z. et al. Swin transformer: Hierarchical vision transformer using shifted windows. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV),* 9992–10002 (IEEE, Montreal, QC, Canada, 2021).
29. Liu, Z. et al. A ConvNet for the 2020s. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR),* 11966–11976 (IEEE, New Orleans, LA, USA, 2022).
30. Ge, Z., Liu, S., Wang, F., Li, Z. & Sun, J. Yolox: Exceeding yolo series in 2021. *arXiv preprint arXiv:2107.08430* (2021).
31. Redmon, J. & Farhadi, A. Yolov3: An incremental improvement. *arXiv preprint arXiv:1804.02767* (2018).
32. Duan, K. et al. CenterNet: Keypoint triplets for object detection. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV),* 6568–6577 (IEEE, Seoul, Korea (South), 2019).
33. Yu, F., Wang, D., Shelhamer, E. & Darrell, T. Deep layer aggregation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition,* 2403–2412 (IEEE, Salt Lake City, UT, USA, 2018).
34. Howard, A. G. et al. Mobilenets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861* (2017).
35. Sandler, M., Howard, A., Zhu, M., Zhmoginov, A. & Chen, L. C. MobileNetV2: Inverted residuals and linear bottlenecks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition,* 4510–4520 (IEEE, Salt Lake City, UT, USA, 2018).
36. Rao, Y. et al. HorNet: Efficient high-order spatial interactions with recursive gated convolutions. In *Advances in Neural Information Processing Systems 35,* 10353–10366 (Neural Information Processing Systems Foundation, Inc. (NeurIPS), NY, USA, 2022).
37. Hu, J., Shen, L. & Sun, G. *IEEE/CVF Conference on Computer Vision and Pattern Recognition,* 7132–7141 (IEEE, Salt Lake City, UT, USA, 2018).
38. Woo, S., Park, J., Lee, J.-Y. & Kweon, I. S. CBAM: Convolutional block attention module. In *Computer Vision – ECCV 2018.* (eds Ferrari, V. et al.) 3–19 (Springer International Publishing, 2018).
39. Cipolla, R., Gal, Y. & Kendall, A. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition,* 7482–7491 (IEEE, Salt Lake City, UT, USA, 2018).
40. Sener, O. & Koltun, V. Multi-task learning as multi-objective optimization. In *Adv. Neural Inf. Process. Syst.* (eds Bengio, S. et al.) (Curran Associates, Inc., 2018).
41. Wang, Y. et al. A simulation method for stellar and target images based on the CODEV optical model. *J. Phys. : Conf. Ser.* **2977**, 012092. https://doi.org/10.1088/1742-6596/2977/1/012092 (2025).

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